Curriculum Vitae

Junge Hyun

Postdoctoral researcher, KIST
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Google scholar: https://scholar.google.com/citations?user=WhhjYQwAAAAJ

Education

Ph.D. (Environmental Science/ Soil Science), Kyung Hee University South Korea
Mar. 2020 - Feb. 2023

- Dissertation: "Soil C sequestration in the anthropogenic land uses: estimation, quantification, and broadening to overall ecosystem services"
- Advisor: Professor Gayoung Yoo
- GPA (4.2/4.3, 36 credits)

M.S. (Environmental Science/ Soil Ecology), Kyung Hee University South Korea
Sep. 2017 - Aug. 2019

- Dissertation: "N_2O emission reduction in agricultural soils - Characteristics of N_2O emission from manure ap plied field and reduction mechanism by biochar application -"
- Advisor: Professor Gayoung Yoo
- GPA (4.15/4.3, 26 credits)

BSc (Environmental Engineering), Kyung Hee University South Korea
Mar. 2012 - Aug. 2017

- GPA (3.77/4.5, 146 credits)
  ※ Completed mandatory military service in the Republic of Korea (2014–2016)

Professional Experience

Postdoctoral Researcher, KIST Seoul, South Korea
Sep. 2026 -

- Center for Climate and Carbon Cycle Research (PI: Chang-Eui Park)

Laboratory Associate, Yale University CT, USA
Aug. 2025 - Aug. 2026

- Department of Ecology and Evolutionary Biology (PI: Eric Slessarev)

Editorial Advisory Board, Global Change Biology Dec. 2024 -

- Serve as a member of the Editorial Advisory Board, reviewing manuscripts and advising editors.

Postdoctoral Researcher, Kyung Hee University Suwon, South Korea
Feb. 2023 – Aug. 2026

- Research: soil carbon dynamic modeling on a national scale, developing low-cost MRV system of soil-based CDR technology, and Life cycle assessment of banking industry
- Co-supervise graduate and undergraduate students in field sampling, data analysis, and manuscript preparation.
- Taught 'Soil Ecology' (Spring 2023)

Visiting Researcher, CESBIO - CNRS Toulouse, France
Host: Dr. Ahmad Al Bitar and Eric Ceschia Jun. 2024 - Jul. 2024

- Drafted an opinion paper titled "How above ground biomass files the carbon pools" targeting submission in s ummer 2025.

Presentation and event planning, COP28 Korea Pavilion Dubai, UAE

Host: KHU, SNU, INRAE, CNRS Dec. 2023
Title: Monitoring and Verification of Ecosystem Carbon Absorption

Participant of Training Course, Cornell University Ithaca, USA
Host: Professor Yiqi Luo Jun. 2023
Title: 6th Training Course on Land Carbon Cycle Modeling

Visiting Researcher, INRAE - AgroParisTech Palaiseau, France
Host: Dr. Claire Chenu Apr. 2023 - May. 2023

- Submitted a manuscript titled "Impact of Enhanced Rock Weathering on Soil Organic Carbon Dynamics through Microbial Activity (submitted on Geoderma in Feb. 2025)"

Visiting Researcher, University of Pennsylvania Philadelphia, USA
Host: Professor Alain F. Plante Apr. 2022 - May. 2022

- Conducted soil thermal analysis and advancing method to differentiate soil carbon pools
- Submitted a manuscript titled "Beyond Total C: Integrative Analysis of Carbon Forms in Urban Soils (In revision)"

Research Assistant & Teaching Assistant, Kyung Hee University Suwon, South Korea
Jul. 2016 - Feb. 2023

Research Funding

Sejong Science Fellowship (2024 - 2028)

- **Role: Principal Investigator**
- **Total Funding: $ 413K**
- Funded by National Research Foundation of Korea
- Project Title: Development of a low-cost, high-efficiency MRV (measurement, reporting, verification) system on soil-based carbon dioxide removal technology

Academic Activity

SCI publications: 11 papers

1. Seo, I., Yoo, G., **Hyun, J**., Son, B., Lee, Y., Im, J., Kim, J., & Jeong, S. (2026) The urban green carbon index (UGCI): A spatial framework for suggesting urban carbon management. PLOS ONE, 21(5), e0347782
2. Bae, J., Seo, I., **Hyun, J**., Park, Y., Jeong, M., Kim, J., ... & Yoo, G. (2026). A spatially rich, temporally coherent soil spectral dataset for soil organic carbon estimation. Scientific Data.
3. **Hyun, J.,** Lee, Y., Sohn, D., Jung, S., Cha, I., & Yoo, G. (2025) Evaluation of the Climate Change Mitigation Potential of Digital-Only Banks and a Quantitative Analysis of their Adaptation Co-benefits. Greenhouse Gases: Science and Technology, 15(6)
4. Park, Y., Kim, Y., **Hyun, J**., Chenu, C., & Yoo, G. (2025) Vulnerability of Soil Organic Carbon in Artificially Constructed Urban Green Spaces: Linking Soil Organic Carbon Fractionation, Microbial Dynamics, and Soil Properties. Geoderma, 117364
5. **Hyun, J.,** Plante, A. F., Bae, J., & Yoo, G. (2025) Beyond Total C: Integrative Analysis of Carbon Forms in Urban Soils. European Journal of Soil Science, 76(2), e70107.
6. Kim, Y., Park, Y., **Hyun, J.,** Seo, I., Jeong, M., Ok, Y., Bae, J., Jung, J., & Yoo, G. (2025) A field-measured dataset for soil organic carbon stock in South Korean grasslands. Scientific Data, 12(1), 691.
7. Kim, Y., **Hyun, J.,** Michelsen, A., Kwon, E. E., & Jung, J. Y. (2024) Key determinants of soil labile nitrogen changes under climate change in the arctic: A meta-analysis of the responses of soil labile nitrogen pools to experimental warming and snow addition. Chemical Engineering Journal, 494, 153066.
8. **Hyun, J.,** & Yoo, G. (2024) Modification of the RothC model to evaluate the inconsistent effect of conservation tillage on SOC stock and a suggestion of a national-scale assessment framework. Science of The Total Environment, 907, 168010.

9. **Hyun, J.,** & Yoo, G. (2023) Effect of high labile biochar on N_2O emission from upland soils: a decision tree analysis and an incubation experiment. Geoderma Regional, 34, e00669.
10. **Hyun, J.,** Kim, Y., Kim, A., Plante, A. F., & Yoo, G. (2022) Ecosystem services-based soil quality index tailored to the metropolitan environment for soil assessment and management. Science of the Total Environment, 820, 153301.
11. Kim, Y., **Hyun, J.,** Yoo, S., & Yoo, G. (2021) The role of biochar in alleviating soil drought stress in urban roadside greenery. Geoderma, 404, 115223.
12. Yoo, G., Lee, Y. O., Won, T. J., **Hyun, J.** G., & Ding, W. (2018) Variable effects of biochar application to soils on nitrification-mediated N2O emissions. Science of the Total Environment, 626, 603-611.

Under Review and In Revision: 1 papers

13. Moon, S., Han, H., Park, Y., Hyun, J., Yun, J., Lim, J., Chun, H., Yoon, S., & Lee, H. (Under review) Contrasting texture-mediated controls on CO_2 and N_2O emissions from arable soils, Pedosphere

In preparation

14. **Hyun, J**.& Slessarev, E. Developing a mineral-informed soil carbon model for better predictions of soil carbon persistence
15. **Hyun, J**., Seo, I., & Yoo, G. Key factors that explain the spatiotemporal variability of soil CO_2 flux in regional, continental, and global scale

Domestic publications (in Korean): 4 papers

16. Park, Y., Kim, Y., **Hyun, J**., Jung, J., & Yoo, G. (2024) Assessing Soil Organic Carbon Stock and Stability in Volcanic Grasslands: Implications for Climate Change Mitigation Potential and Management Levels. Korean Journal of Soil Science and Fertilizer, 56(4), 342
17. **Hyun, J.,** Kim, S., Jung, W., Kim, J., & Yoo, G. (2021) Quantifying the Effects of Agricultural Management Practices on Soil Carbon Storage in an Agricultural Environment Conservation Program. Journal of Climate Change Research, 12(6), 701
18. Choi, H., **Hyun, J.,** Kim, Y., & Yoo, G. (2019) Improvement of Ammonia Emission Inventory Estimation Methodology for Fertilizer Application in the Agricultural Sector. Journal of Climate Change Research, 10(3), 237
19. **Hyun, J.,** Yoo, S., Yang, X., Lee, J., & Yoo, G. (2017) Annual Variability in Nitrous Oxide Emission from Agricultural Field Soils. Journal of Climate Change Research, 8(4), 305

Reviewing for Journal: Total 16 reviews in 2024 - 2026

20. Global Change Biology: 9 papers
21. Journal of Environmental Management: 2 papers
22. Science of the Total Environment: 1 paper
23. Environmental Monitoring and Assessment: 1 paper
24. Geoderma Regional: 1 paper
25. Resources, Environment and Sustainability: 1 paper
26. Soil & Tillage Research: 1 paper

International conferences

27. Park, Y., Ok, Y., **Hyun, J.,** Seo, I., & Yoo, G. (2025). Soil Organic Carbon (SOC) Fraction Dynamics Influenced by the Enhanced Rock Weathering with Blast Furnace Slag and Mine Tailings. In EGU General Assembly Conference Abstracts (pp. EGU25-19373).
28. Ok, Y., **Hyun, J.,** Park, Y., & Yoo, G. (2024) Assessing the Impact of Enhanced Rock Weathering on Soil Biological Interactions with Comprehensive Carbon Dioxide Removal Index. EGU24 General Assembly
29. Park, Y., **Hyun, J**., Ok, Y., & Yoo, G. (2024) Impact of Enhanced Rock Weathering on Soil Organic Carbon Dynamics through Microbial Activity. EGU24 General Assembly
30. **Hyun, J.,** Bae, J., & Yoo, G. (2023) Assessment of thermal analysis techniques for determining organic, black, and inorganic carbon contents in urban soils. EGU23 General Assembly

31. **Hyun, J.,** & Yoo, G. (2022) Feasibility of 4 per 1000 initiative analyzed from national scale assessment of SOC sequestration by conservation tillage practices. AGU Fall meeting 2022
32. **Hyun, J.,** Kim, Y., Kim, A., Plante, A. F., & Yoo, G. (2021) A soil quality index tailored to metropolitan environment for assessment and management policy development AGU Fall meeting 2021

Patents

1. Yoo, G., **Hyun, J.**, & Kim, S. (2025). A method and system for calculating variation quantitatively of Soil Carbon storage in agricultural. KR 10-2840903, South Korea
2. Yoo, G. & **Hyun, J.** (2024). Soil organic carbon storage change prediction system and prediction method by tillage method. KR 10-2710164, South Korea
3. Yoo, G., **Hyun, J.**, & Kim, Y. (2023). Ecosystem services-based soil quality index tailored to the metropolitan environment for soil assessment and management. KR 10-2512415, South Korea
4. Yoo, G., Seo, I., Jeong, M., & **Hyun, J.** (2022). Planting module for Performance Enhancement of Carbon Absorption and Storage Amount and Constructing Method thereof. KR 10-2352273, South Korea

Participated Project (Recent 5 year)

1. Development of a low-cost, high-efficiency MRV system on soil-based CDR technology (2024-, National Research Foundation of Korea)
2. Korea Carbon Project: Development of a technology for an observation-based GHG emissions geospatial information map platform (2023-, Ministry of Environment)
3. Digital mapping on carbon storage and absorption for Gyeonggi province (2024-2025, Gyeonggi province)
4. Development of a Tier 3 level methodology for estimating cropland soil C stock change (2023-, Rural Development Administration)
5. Kakaobank green value report (2023-2024, KakaoBank)
6. Monitoring soil CO_2 emission in urban forests (2022-2025, Korea Forest Service)

Additional projects available upon request

Technical Skills

Modeling and Assessment

- Soil Carbon dynamic modeling (Assessment of the effect of human activity on SOC)
- Data-driven model with machine learning algorithms (random forest, ANN, LGBM)
- Calibration and parameter optimization of model: Multi-objective optimization of process-based model with genetic algorithm, Bayesian optimization of data-driven model
- Life-cycle assessment (LCA)
- Meta-analysis

Programming skills

- R & Python: Basic statistics, process-based model, and machine learning algorithms (+AI-assisted development)
- QGIS: Geographic information processing
- Google Earth Engine (JavaScript): Geographic information processing

Physical, Chemical, and Biological Analysis of Soil

- **Soil physical properties** (Bulk density, texture, resistance, Size and density fractionation, etc.)
- **Soil chemical properties** (pH, EC, Total C and N content (EA), Dissolved organic C (TOC analyzer), Cation exchange capacity, Inorganic N content (AA3), etc.)
- **Soil biological properties** (Soil DNA extraction and quantitative analysis by real-time PCR, Microbial Biomass Nitrogen and Carbon (Chloroform fumigation), Enzyme activity(Hydrolase and oxidase), Gas flux (CO_2, CH_4, and N_2O)

Instruments: Gas Chromatography, Soil CO_2 flux system (Li-8250A and EGM-5), NIR Spectrometer, Elemental Analyzer, NanoDrop, real-time PCR